

Project 12

Nonsingular Posture Change on custom_3R

Format: Group of two students; simulation-based project.

Course foundations: Lectures 3, 5–7; Exercise 3.

Motivation

For a cuspidal robot, two distinct configurations can reach the same workspace position while belonging to one singularity-free configuration region. A nonsingular connecting joint path then produces a closed tool path. This explains why posture labels and endpoint positions alone are insufficient to understand global motion planning.

Task

Overall objectives. Use the exact custom_3R model to construct and validate a candidate nonsingular connection between two distinct IK solutions of one position, and interpret both its joint-space path and its closed workspace image.

- **Validate the exact model.** Read the URDF and establish its D–H parameters, angle offsets, and selected operational point. Compare forward kinematics and Jacobians with the course implementation. The exact robot is orthogonal with non-intersecting consecutive axes; do not substitute the idealized atlas dimensions.
- **Locate the IKs on reduced maps.** Plot $\det J_p$ and its zero set in (q_2, q_3) , then map the critical curves and two-/four-IK regions in (ρ, z) . Select a regular target and overlay all its distinct IKs. Identify two candidate configurations in the same computed singularity-free component; equal determinant signs alone are insufficient.
- **Construct the joint-space connection.** Use the Lecture 6 nonsingular-change demonstration to obtain a starting candidate, then construct or adjust a piecewise-smooth path in the reduced component. Specify a continuous q_1 interpolation connecting the full endpoint IKs. Recover and display the complete three-joint trajectory.
- **Validate the workspace loop.** Compute the forward image of every joint sample and check that the endpoint positions agree while endpoint configurations are distinct modulo 2π . Display both the three-dimensional workspace loop and its meridian projection. Do not infer full Cartesian closure from agreement in (ρ, z) alone.
- **Test regularity and robustness.** Monitor $\det J_p$, the smallest singular value, and optional joint-limit violations. Refine near the smallest margin and compare the original path with one small path perturbation. Separate strong sampled evidence from a continuous mathematical guarantee; use an analytical bound only if readily available.

Specifics and scope

Study one target, one selected IK pair, and two candidate paths (the original and one variation). The lecture's existing connection is an allowed seed, but validation must use the exact model. The primary cuspidality study uses periodic unconstrained joints; report separately whether the same path respects the URDF limits. No obstacle model or general cusp-search algorithm is required.

Use the position Jacobian J_p for the 3R task. Confirm that the meridian reduction is valid for the selected model and operational point. Count distinct real IKs modulo periodicity, then filter joint limits separately. Identify two-/four-IK regions where present and explain any absent count rather than forcing both types into every map. Show both determinant values and the zero contour, and distinguish sampled connectivity from a proof. Refine the decisive region instead of claiming completeness from a coarse grid.

Expected outcome

Explain cuspidality through a concrete nonsingular posture change, including the relationship between reduced singularities, multiple IKs, and a closed workspace image. Deliver the selected IK pair, reproducible joint paths, paired plots, and a regularity/closure validation report.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The exact custom_3R is an orthogonal 3R robot with non-intersecting consecutive joint axes. Its URDF is `lectures_main/assets/models/custom_3R/custom_3R_new.urdf`; STL meshes are in the same directory. Use the lecture website to inspect axes and frames, verify D–H parameters, plot IKs, and animate configurations. Use the exact-model nonsingular-change lab in Lecture 6 and the Lecture 3 IK method. The idealized atlas can explain concepts but must not supply metric results for this exact-URDF experiment.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL's 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.